MODULE 3 BACK AND THORAX MUSCLES

Muscles of the Back and Thorax.

TRUNK MUSCULATURE

The muscles that move the scapula, extend the spine, and build the thoracic wall. Work these in layers, superficial to deep, because that is how they are dissected and how they are tagged in lab.

BY THE END

1. Identify the superficial and deep muscles of the back and state the action of each.
2. Name the three columns of the erector spinae and their order from the midline out.
3. Identify the muscles of the thoracic wall and the direction their fibers run.
4. Describe the diaphragm, its openings, and what passes through each.
5. Name the anterior thoracic muscles that act on the scapula and the humerus.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Back and thorax

drsrennie-stack.github.io/new-build-bio4-solano/back-thorax-concept-videos.html

Superficial Back

The superficial layer is scapular. Every muscle here crosses from the axial skeleton to the pectoral girdle or the humerus, so every action is a scapular action or an arm action.

SUPERFICIAL MUSCLES OF THE BACK

Muscle	Origin	Insertion	Action
Trapezius	Occipital bone, ligamentum nuchae, spinous processes C7 to T12	Lateral clavicle, acromion, spine of the scapula	Upper fibers elevate the scapula, middle retract, lower depress; all rotate the glenoid upward
Latissimus dorsi	Spinous processes T7 to L5, thoracolumbar fascia, iliac crest, lower ribs	Intertubercular sulcus of the humerus	Extends, adducts, and medially rotates the arm
Levator scapulae	Transverse processes C1 to C4	Superior medial border of the scapula	Elevates the scapula and rotates the glenoid downward
Rhomboid major	Spinous processes T2 to T5	Medial border of the scapula, below the spine	Retracts the scapula and rotates the glenoid downward
Rhomboid minor	Spinous processes C7 to T1	Medial border of the scapula, at the spine	Retracts the scapula and rotates the glenoid downward

HOLD ONTO THIS

Trapezius and the rhomboids both retract, and they are direct antagonists in rotation: trapezius turns the glenoid up, the rhomboids turn it down. That pairing is a favourite exam question because the retraction looks like agreement.

Deep Back

The erector spinae

Three vertical columns running the length of the back. Name them in order from the midline outward, because that order is how they are identified on a cadaver and on a cross section.

THE THREE COLUMNS, LATERAL TO MEDIAL

Column	Position	Extent
Iliocostalis	The most lateral column	Attaches to the ribs, in lumbar, thoracic, and cervical parts
Longissimus	The intermediate column	The longest column, reaching to the mastoid process
Spinalis	The most medial column	Runs along the spinous processes, smallest of the three

DEEPER AND ASSOCIATED MUSCLES OF THE BACK

Muscle	Origin	Insertion	Action
Splenius capitis	Ligamentum nuchae, spinous processes C7 to T3	Mastoid process and occipital bone	Together extend the head; singly rotate the head to the same side
Splenius cervicis	Spinous processes T3 to T6	Transverse processes C1 to C3	Extends and rotates the neck
Semispinalis	Transverse processes C7 to T12	Occipital bone and spinous processes above	Extends the head and vertebral column
Multifidus	Sacrum, ilium, transverse processes	Spinous processes two to four segments above	Extends and rotates the vertebral column, and stabilises it segment by segment
Quadratus lumborum	Iliac crest and iliolumbar ligament	Rib 12 and transverse processes L1 to L4	Laterally flexes the trunk; both together extend the lumbar spine

HOLD ONTO THIS

Erector spinae is one name for three columns, and the mnemonic is I Love Spaghetti: Iliocostalis, Longissimus, Spinalis, lateral to medial.

The Thoracic Wall

MUSCLES OF THE THORACIC WALL

Muscle	Origin	Insertion	Action
External intercostals	Inferior border of the rib above	Superior border of the rib below	Elevate the ribs. Fibers run downward and forward, hands in pockets
Internal intercostals	Superior border of the rib below	Inferior border of the rib above	Depress the ribs. Fibers run at right angles to the external layer

Muscle	Origin	Insertion	Action
Innermost intercostals	Internal surface of the rib below	Internal surface of the rib above	Act with the internal intercostals; separated from them by the neurovascular bundle
Transversus thoracis	Posterior surface of the lower sternum	Costal cartilages 2 to 6	Depresses the ribs
Serratus posterior superior	Spinous processes C7 to T3	Ribs 2 to 5	Elevates the ribs
Serratus posterior inferior	Spinous processes T11 to L2	Ribs 9 to 12	Depresses the ribs

The diaphragm

The diaphragm is the floor of the thoracic cavity and the chief muscle of quiet breathing. Its openings are as testable as the muscle itself.

THE DIAPHRAGM AND ITS PARTS

Structure	What it is
Origin	The xiphoid process, the inner surface of ribs 7 to 12, and the lumbar vertebrae by two crura
Insertion	The central tendon, a flat aponeurosis at its own dome
Action	Contracts and flattens, which increases the vertical dimension of the thoracic cavity
Nerve supply	The phrenic nerve, from spinal segments C3, C4, and C5
Caval opening	At vertebral level T8, transmits the inferior vena cava and the right phrenic nerve
Esophageal hiatus	At vertebral level T10, transmits the esophagus and the vagus nerves
Aortic hiatus	At vertebral level T12, transmits the aorta, thoracic duct, and azygos vein

HOLD ONTO THIS

The three openings sit at even levels, 8, 10, 12, and the contents follow the count: vena cava at 8, esophagus at 10, aorta at 12. For the nerve supply, C3, 4, and 5 keep the diaphragm alive.

Anterior Thorax

ANTERIOR THORACIC MUSCLES ACTING ON THE GIRDLE AND THE ARM

Muscle	Origin	Insertion	Action
Pectoralis major	Medial clavicle, sternum, costal cartilages 1 to 6	Lateral lip of the intertubercular sulcus of the humerus	Flexes, adducts, and medially rotates the arm
Pectoralis minor	Ribs 3 to 5	Coracoid process of the scapula	Depresses and protracts the scapula, and elevates the ribs when the scapula is fixed

Muscle	Origin	Insertion	Action
Serratus anterior	Ribs 1 to 8 or 9, on the lateral chest wall	Anterior surface of the medial border of the scapula	Protracts the scapula and rotates the glenoid upward; holds the scapula against the chest wall
Subclavius	Rib 1 at the costal cartilage	Inferior surface of the clavicle	Depresses and steadies the clavicle

WHERE THESE MUSCLES SHOW UP IN THE CLINIC

The **serratus anterior** is supplied by the long thoracic nerve, which runs superficially down the chest wall and is exposed in axillary surgery. Lose the nerve and the scapula lifts off the back when the patient pushes against a wall, a winged scapula. The muscle's job, holding the scapula flat, is visible in its absence.

The **diaphragm** is supplied from C3, C4, and C5, high in the neck rather than at the level of the muscle itself, because the diaphragm develops in the neck and descends. A cervical spinal cord injury above C3 stops breathing; below C5 it does not.

Referred pain follows the same nerve. Irritation under the diaphragm, from blood or infection, is felt in the shoulder tip, because the **phrenic nerve** and the skin of the shoulder share those cord segments.

The **latissimus dorsi** brings the arm down and back, which is why it powers the movement of pushing up out of a chair and why it is called the swimmer's muscle.